

constraints, this is identical to the one-period mean-variance problem that we covered in chapter 3. The solution for a single risky asset with expected return μ and volatility σ , with a risk-free rate of r_f is given in equation (4.5), and we denote it by x_{t+4}^* , where the asterisk means that the portfolio weight is optimal. The investor holds x_{t+4}^* in equities and $(1 - x_{t+4}^*)$ in risk-free bonds. In principle, this portfolio weight can depend on what expected return and volatility are prevailing at $t+4$ (say, the economy is booming or in bust).

The maximum utility obtained at $t+4$ is:

$$V_{t+4} = E[U(1 + r_{p,t+5}^*)], \quad (4.9)$$

where the portfolio return from $t+4$ to $t+5$, $r_{p,t+5}^*$, is a function of the optimal portfolio weight chosen at $t+4$, x_{t+4}^* , so $r_{p,t+5}^* = r_{p,t+5}(x_{t+4}^*)$. The maximum utility V_{t+4} in equation (4.9) is called the *indirect utility*, and it potentially differs across economic conditions prevailing at time $t+4$.

Having solved the last period's problem, let us turn to the problem two periods before the end. At $t+3$, we need to solve both the portfolio weights at $t+3$ and $t+4$, which are x_{t+3} and x_{t+4} , respectively:

$$\begin{aligned} \max_{\{x_{t+3}, x_{t+4}\}} & U(W_{t+3})E[U(1 + r_{p,t+4}(x_{t+3}))U(1 + r_{p,t+5}(x_{t+4}))] \\ &= \max_{\{x_{t+3}, x_{t+4}\}} E[U(1 + r_{p,t+4}(x_{t+3}))U(1 + r_{p,t+5}(x_{t+4}))]. \end{aligned} \quad (4.10)$$

But, we already solved the last period problem and found the optimal portfolio weight at $t+4$, x_{t+4}^* for any outcome at $t+3$. This allows us to write the problem two periods before the end as the problem from $t+3$ to $t+4$, plus the problem with the known solution that we solved from $t+4$ to $t+5$:

$$\begin{aligned} \max_{\{x_{t+3}, x_{t+4}\}} & E[U(1 + r_{p,t+4}(x_{t+3}))U(1 + r_{p,t+5}(x_{t+4}))] \\ &= \max_{x_{t+3}} E[U(1 + r_{p,t+4}(x_{t+3}))U(1 + r_{p,t+5}(x_{t+4}^*))] \\ &= \max_{x_{t+3}} E[U(1 + r_{p,t+4}(x_{t+3}))V_{t+4}]. \end{aligned} \quad (4.11)$$

The first equality in equation (4.11) substitutes the last period's solution into the time $t+3$ problem. This now leaves just one portfolio weight at $t+3$, x_{t+3} , to solve. The second equality says that this problem is a standard single-period problem, except that it involves the indirect utility V_{t+4} , but we know everything about the indirect utility and the optimal strategies at $t+4$ from solving the last period's problem (equation (4.9)). We can solve the problem in equation (4.11) as a one-period problem, and we denote the optimal weight at $t+3$ as x_{t+3}^* . It has the same as the one-period solution in equation (4.5), except that we adjust equation (4.5) for the optimized strategies adopted at $t+4$ that are captured by the indirect utility, V_{t+4} . Panel B of Figure 4.2 shows this pictorially. Given the known solution at $t+4$, we use the optimal portfolio weight at $t+4$ to solve the portfolio weight at